# Practice Quiz: Body Science — Module 07: Communication

**Name**: ___ *Date*: ___

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## Part A: Multiple Choice

1. Neurons communicate with each other by:

A) Talking out loud B) Sending electrical signals and releasing chemical messengers (neurotransmitters) across tiny gaps called synapses C) Touching each other directly D) Using Wi-Fi

1. Hormones are chemical messengers that travel through your blood. Compared to nerve signals, hormones:

A) Are faster and more targeted B) Are slower but affect many parts of the body at once — like a broadcast message rather than a direct text C) Don't carry information D) Only exist in the brain

1. When your body detects an infection, immune cells release signaling molecules called cytokines. This is an example of:

A) Cells being random B) Communication between agents — immune cells sharing information about a threat so other cells can respond C) Something that has nothing to do with communication D) Only the brain being involved

1. The gut-brain axis is the communication pathway between your digestive system and your brain. This shows that:

A) Your gut and brain never talk to each other B) Different body systems communicate constantly to coordinate their predictions and actions C) Only the brain sends signals to the gut, never the other way around D) Communication in the body only uses nerves

1. When you blush with embarrassment, your face turns red even though you didn't choose to blush. This happens because:

A) Your skin is malfunctioning B) Your autonomic nervous system communicated the emotional state to your blood vessels — an involuntary signal that others can perceive C) Blushing is completely voluntary D) Only certain people blush

1. Your endocrine system (hormones) and your nervous system both carry messages, but in different ways. Together, they show that:

A) The body has only one communication method B) The body uses multiple communication channels that work at different speeds and scales to coordinate its systems C) Hormones and nerves are unrelated D) Communication isn't important for the body

1. When a mother hears her baby cry, her body releases oxytocin, which triggers milk production. This is an example of:

A) A random body response B) Communication between agents — the baby's cry is a signal that the mother's body interprets and acts on C) Something that has nothing to do with predictions D) A response that requires no perception

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## Part B: Short Answer

1. Compare how **neurons** communicate (fast, targeted, electrical and chemical) with how **hormones** communicate (slower, widespread, chemical). Give an example of a situation where each type is useful. How is this like the difference between sending a text message versus making an announcement over a loudspeaker?

2. Explain how your **immune system communicates** internally when it detects a virus. What signals are sent? How do other immune cells "get the message"? How does this coordination help your body respond more effectively?

3. Describe the **gut-brain axis** in your own words. How does your gut communicate with your brain, and how does your brain communicate back? Give an example of when you "felt something in your gut" and explain what might have been happening biologically.